

**Course: Full Stack Web Development**

**Course Code: 23CM4121**

**Experiment No.: 5**

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## **TITLE**

Develop Dynamic Webpage Using JavaScript

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## **Aim**

To develop a dynamic webpage using JavaScript that accepts student details and subject marks, calculates the total marks, average, grade, and result, and dynamically displays the results on the webpage.

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## **OBJECTIVE**

The objectives of this experiment are:

- To understand the basics of JavaScript and DOM manipulation.
  - To accept student details and subject marks through HTML input fields.
  - To retrieve input values using JavaScript.
  - To calculate total and average marks.
  - To determine the grade based on the average marks.
  - To determine whether the student has passed or failed.
  - To dynamically update the webpage using JavaScript.
  - To implement a Reset button for clearing the entered data.
  - To apply dynamic styling based on the grade.
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## **Theory**

JavaScript can be used to make a webpage dynamic and interactive by responding to user actions and modifying HTML elements without reloading the page.

In this experiment, a Student Marks Calculator is developed using HTML, CSS, and JavaScript. The webpage accepts the student's name and marks in five subjects: English, Maths, Science, Social, and

SL.

When the Calculate button is clicked, JavaScript retrieves the values entered by the user and calculates the total and average marks. Based on the average, a grade is assigned from A to D or Fail. The program also checks whether the student has scored at least 35 marks in every subject to determine whether the student has passed or failed.

The webpage uses DOM manipulation and event listeners to dynamically update the result section. The grade color is also changed dynamically according to the grade. The Reset button clears all the entered values and resets the displayed results.

### Main Features

- Accepts student's name.
- Accepts marks for five subjects.
- Calculates total marks.
- Calculates average marks.
- Assigns a grade based on average.
- Determines PASS or FAIL.
- Changes the grade color dynamically.
- Displays results without refreshing the webpage.
- Provides a Reset button to clear the form.

### Concepts Used

| JavaScript Concept | Purpose                              |
|--------------------|--------------------------------------|
| getElementById()   | Accesses HTML elements               |
| addEventListener() | Handles button click events          |
| Number()           | Converts input values into numbers   |
| if-else            | Determines grade and result          |
| textContent        | Dynamically changes webpage text     |
| .value             | Gets or modifies input values        |
| DOM Manipulation   | Updates webpage elements dynamically |
| style.color        | Changes the grade color dynamically  |

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## ALGORITHM / PROCEDURE

1. Create an HTML webpage containing input fields for the student's name and five subject marks.
2. Create Calculate and Reset buttons.

3. Create a result section to display the student name, total, average, grade, and result.
4. Apply CSS styling to the webpage and its components.
5. Create a JavaScript file and select the required HTML elements using `getElementById()`.
6. Add a click event listener to the Calculate button.
7. Retrieve the student's name and subject marks using `.value`.
8. Convert the marks into numbers using `Number()`.
9. Calculate the total marks.
10. Calculate the average by dividing the total by 5.
11. Determine the grade using if-else conditions.
12. Change the grade color according to the grade.
13. Check whether all subjects have at least 35 marks.
14. Display PASS if all subjects satisfy the minimum marks requirement; otherwise display FAIL.
15. Add a click event listener to the Reset button.
16. Clear all input fields and reset the displayed results.
17. Execute the application and verify the output using multiple test cases.

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## Code

### HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student grade calculator</title>
  <link rel="stylesheet" href="./style.css">
</head>
<body>
  <div>
    <div class="heading">
      <h1>Student Grade calculator</h1>
    </div>

    <div class="marks">
      <label for="" class="name">Name</label>
      <input type="text" placeholder="Enter the your name" class="ans" id="studentName">

      <br><br>
      <label for="" class="name">English</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="eng">

      <br><br>
      <label for="" class="name">Maths</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="math">

      <br><br>
      <label for="" class="name">Science</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="sci">

      <br><br>
      <label for="" class="name">Social</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="soc">

      <br><br>
      <label for="" class="name">Sl</label>
      <input type="number" placeholder="Enter the marks" class="ans" id="sl">

      <br><br>

      <button class="btn" id="calculateBtn">Calculate</button>
      <button type="reset" class="btn" id="resetBtn">Reset</button>

    </div>

    <div class="results">
      <h3>Results: </h3>
      <p id="student">Name :</p>
      <p id="total">Total :</p>
      <p id="average">Average: </p>
      <p id="grade">Grade: </p>
      <p id="result">Result: </p>
    </div>
  </div>

  <script src="./script.js"></script>
</body>
</html>
```

## CSS Code:

```
body{
  /* display: flex; */
  background-color: blanchedalmond;
}
/* .heading{
  display: flex;
  justify-content: center;
} */
h1{
  /* display: flex; */
  /* justify-content: center; */
  display: flex;
  justify-content: center;
  margin: 0 auto 20px auto;
  /* border: 2px solid black; */
  background-color: white;
  width: 500px;
  flex-wrap: wrap;
  border-radius: 20px;
  padding: 10px;
  box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.2);
}

.marks{
  background-color: rgb(199, 167, 230);
  margin: 80px;
  padding: 40px;
  border-radius: 15px;
  /* gap: 40px; */
  display: flex;
  flex-direction: column;
  justify-content: center;
  /* text-align: center; */
  gap: 10px;
}

.name{
  /* border: 2px solid black; */
  /* margin: 0 auto 10px auto; */
  color: rgb(19, 17, 173);
  font-size: 1rem;
}

.ans{
  border-radius: 5px;
  border-color: none;
  margin: 10px;
  padding: 8px;
  width: 250px;
}

.btn{
  width: 100px;
  display: flex;
  border-color: none;
  background-color: rgb(106, 106, 226);
  border-radius: 8px;
  color: white;
  padding: 8px;
  box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.15);
}

.btn:hover{
  transform: scale(1.2);
  transition: 1s;
}
```

```
.results h3{
  color: white;
  display: flex;
  justify-content: center;
  font-size: 1.5rem;
}

.results p{
  color: rgb(72, 64, 190);
  font-weight: bold;
}
```

# Javascript code:

```
const calculateBtn = document.getElementById("calculateBtn");
calculateBtn.addEventListener("click", function(){

const studentName = document.getElementById("studentName").value;
document.getElementById("student").textContent = `Name : ${studentName}`

const english = Number(document.getElementById("eng").value);
const maths = Number(document.getElementById("math").value);
const science = Number(document.getElementById("sci").value);
const social = Number(document.getElementById("soc").value);
const sl = Number(document.getElementById("sl").value);

const total = english + maths + science + social + sl;
console.log(total);

const totalElement = document.getElementById("total");
totalElement.textContent = `Total : ${total}`;

const average = total / 5;
const avgElement = document.getElementById("average");
avgElement.textContent = `Average: ${average}`;

let grade;
if(average >= 90){
    grade = "A";
}
else if(average >= 80){
    grade = "B";
}
else if(average >= 70){
    grade = "C";
}
else if(average >= 60){
    grade = "D";
}
else{
    grade = "Fail";
}

document.getElementById("grade").textContent = `grade : ${grade}`;

const gradeElement = document.getElementById("grade");
if(grade == "A"){
    gradeElement.style.color = "green";
}
else if(grade == "B"){
    gradeElement.style.color = "blue";
}
else if(grade == "C"){
    gradeElement.style.color = "yellow";
}
else if(grade == "D"){
    gradeElement.style.color = "orange";
}
else{
    gradeElement.style.color = "red";
}

let result;

if (
    english >= 35 &&
    maths >= 35 &&
    science >= 35 &&
    social >= 35 &&
    sl >= 35
) {
    result = "PASS";
}
else {
    result = "FAIL";
}
document.getElementById("result").textContent = `Result : ${result}`;
});

const resetBtn = document.getElementById("resetBtn");
resetBtn.addEventListener("click", function () {
    document.getElementById("studentName").value = "";
    document.getElementById("eng").value = "";
    document.getElementById("math").value = "";
    document.getElementById("sci").value = "";
    document.getElementById("soc").value = "";
    document.getElementById("sl").value = "";

    document.getElementById("student").textContent = "Name :";
    document.getElementById("total").textContent = "Total :";
    document.getElementById("average").textContent = "Average :";
    document.getElementById("grade").textContent = "Grade :";
    document.getElementById("result").textContent = "Result :";

    document.getElementById("grade").style.color = "";
});
```

```
let result;

if (
    english >= 35 &&
    maths >= 35 &&
    science >= 35 &&
    social >= 35 &&
    sl >= 35
) {
    result = "PASS";
}
else {
    result = "FAIL";
}
document.getElementById("result").textContent = `Result : ${result}`;
});

const resetBtn = document.getElementById("resetBtn");
resetBtn.addEventListener("click", function () {
    document.getElementById("studentName").value = "";
    document.getElementById("eng").value = "";
    document.getElementById("math").value = "";
    document.getElementById("sci").value = "";
    document.getElementById("soc").value = "";
    document.getElementById("sl").value = "";

    document.getElementById("student").textContent = "Name :";
    document.getElementById("total").textContent = "Total :";
    document.getElementById("average").textContent = "Average :";
    document.getElementById("grade").textContent = "Grade :";
    document.getElementById("result").textContent = "Result :";

    document.getElementById("grade").style.color = "";
});
```

## CODE EXPLANATION

| Code                                    | Explanation                                                    |
|-----------------------------------------|----------------------------------------------------------------|
| getElementById()                        | Selects an HTML element using its ID.                          |
| addEventListener()                      | Handles events such as button clicks.                          |
| .value                                  | Retrieves the value entered in an input field.                 |
| Number()                                | Converts the entered marks from string values into numbers.    |
| english + maths + science + social + sl | Calculates the total marks obtained in all five subjects.      |
| total / 5                               | Calculates the average marks.                                  |
| if-else                                 | Determines the grade and pass/fail result based on conditions. |

|                             |                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------|
| textContent                 | Dynamically updates the result displayed on the webpage.                         |
| style.color                 | Changes the grade color dynamically according to the grade.                      |
| >= 35                       | Checks whether the student has scored the minimum passing marks in each subject. |
| gradeElement                | Stores the HTML element used to display the grade.                               |
| resetBtn                    | Selects the Reset button from the webpage.                                       |
| .value = ""                 | Clears the values entered in the input fields.                                   |
| innerText                   | Updates the displayed text such as the total number of contacts or results.      |
| style.color = ""            | Resets the dynamically applied grade color.                                      |
| resetBtn.addEventListener() | Handles the click event of the Reset button.                                     |

## TEST CASES AND EXECUTION DATA

| Test Case | Component Tested            | Action / Input                                                             | Expected Result                                          | Actual Result                           | Status |
|-----------|-----------------------------|----------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------|--------|
| TC-01     | Calculate Total and Average | Enter Name: Rahul and marks: 80, 90, 85, 75, 70 and click <b>Calculate</b> | Total and average should be calculated correctly         | Total and average displayed correctly   | Pass   |
| TC-02     | Grade Calculation           | Enter marks with an average of 90 or above                                 | Grade should be <b>A</b> and displayed in green          | Grade A displayed in green              | Pass   |
| TC-03     | Pass/Fail Result            | Enter marks of at least 35 in all five subjects                            | Result should be <b>PASS</b>                             | PASS displayed successfully             | Pass   |
| TC-04     | Fail Result                 | Enter less than 35 marks in any one subject                                | Result should be <b>FAIL</b>                             | FAIL displayed successfully             | Pass   |
| TC-05     | Reset Function              | Enter student details, calculate the result, and click <b>Reset</b>        | All input fields and displayed results should be cleared | Fields and results cleared successfully | Pass   |

## Output :

Student Grade calculator

Name

English

Maths

Science

History

Art

Calculate

Reset

Enter Name

Total: 420

Average: 84.0

grade: B

Result: 30.00

Results:

## RESULT

Thus, a dynamic Student Marks Calculator was successfully developed using HTML, CSS, and JavaScript. The application successfully accepts student details and subject marks, calculates the total and average, determines the grade and PASS/FAIL result, dynamically changes the grade color, and resets the form using JavaScript DOM manipulation and event handling.